

DAC Clocking and I2S Interface (ES9038Q2M): Working Document

Digital Audio Player • Rev 9

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1. Summary

Field	Value
Manufacturer / MPN	ESS Technology, ES9038Q2M
Package	30-lead QFN with exposed pad (exposed pad = DGND)
Function	32-bit stereo audio DAC, differential current outputs
Timing role	I2S master. The ESP32-S3 is I2S slave.
Clock sources	Two tri-state oscillators on XI: 22.5792 MHz (44.1 kHz family) and 24.576 MHz (48 kHz family)

The DAC is the I2S timing master so that its DPLL can be turned off (synchronous mode). In this mode every audio clock, including the DAC's internal MCLK, BCLK and LRCLK, is derived from one local low-noise oscillator, which gives the best audio performance. The ESP32-S3 only supplies PCM data.

2. Design Requirements

Requirement	Value
Input format	PCM over I2S. No DSD, no S/PDIF.
Sample rates	44.1 kHz family: 44.1, 88.2, 176.4 kHz. 48 kHz family: 48, 96, 192 kHz.
Timing roles	DAC master (Reg 10 master_mode = 1), ESP32-S3 slave
DPLL	Off, synchronous mode (Reg 10 128fs_mode = 1): MCLK = 128 × FSR exactly
Clocking	Two external tri-state oscillators on XI, one per family, one enabled at a time
XI frequency	≤ 100 MHz. Chosen for lowest phase noise.

Requirement	Value
Maximum sample rate	176.4 / 192 kHz, fixed by the selected oscillators. 352.8 / 384 kHz would need 45.1584 and 49.152 MHz oscillators.
I2C speed	100 kHz standard mode (SCL < MCLK / 20)
Clock switching	Only while the DAC is muted or RESETB is low
Constraints	Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1	AVCC_R	+3V3_ANALOG, 0.1 μ F + 1 μ F to GND
2	DACRB	Right negative current output, to I/V stage
3	DACR	Right positive current output, to I/V stage
4	BIAS	Not connected
5	SW	Not connected
6	FSYNC	Not connected
7	HPSDb	Not connected
8	DACL	Left positive current output, to I/V stage
9	DACLB	Left negative current output, to I/V stage
10	AVCC_L	+3V3_ANALOG, 0.1 μ F + 1 μ F to GND
11	AGND_L	GND
12	VCCA	+3V3_ANALOG, 0.1 μ F + 1 μ F to GND
13	DVCC	+3V3_DIG, 0.1 μ F to GND
14	DGND	GND
15	DVDD	4.7 μ F to GND only (internal 1.2 V core regulator output)
16	GPIO2	ESP32-S3 GPIO, 10 k Ω pull-down (see 3.1)
17	GPIO1	ESP32-S3 GPIO, 10 k Ω pull-down (see 3.1)
18	DATA2	PCM data input from ESP32-S3
19	DATA1	LRCLK output to ESP32-S3
20	DATA_CLK	BCLK output to ESP32-S3
21	RESETB	ESP32-S3 GPIO, 10 k Ω pull-down
22	RT1	GND (reserved, must connect to DGND)
23	SCL	ESP32-S3, 10 k Ω pull-up to +3V3_DIG
24	SDA	ESP32-S3, 10 k Ω pull-up to +3V3_DIG
25	ADDR	10 k Ω pull-up to +3V3_DIG (I2C address 0x92)
26	NC	Not connected
27	VCCA	+3V3_ANALOG, 0.1 μ F + 1 μ F to GND
28	XOUT	Not connected (XI driven by an oscillator)
29	XI	Oscillator outputs, each through a 22 Ω series resistor
30	AGND_R	GND
EP	Exposed pad	DGND (GND)

Source: [ES9038Q2M Datasheet v1.4 \(ESS Technology\)](#), pp. 4 to 5 (pin descriptions); p. 14 (I2C address).

3.1 GPIO1 and GPIO2 Functions

Each GPIO is set independently in Register 8 (gpio1_cfg bits 3:0, gpio2_cfg bits 7:4). The default is mode 13, an input that shuts the DAC down when pulled high, so both pins carry 10 k Ω pull-downs to keep the DAC running before firmware configures them.

Mode	Name	Direction	Behaviour
0	Automute Status	Output	High when automute is triggered
1	Lock Status	Output	High when the DAC is locked to its clocks
2	Volume Min	Output	High when all volume controls have ramped to minus full scale
3	CLK	Output	Buffered MCLK for synchronising other devices
4	Automute/Lock Interrupt	Output	High when lock or automute status changes; cleared by reading Register 64
5	Amplifier_PDB	Output	Follows Register 39 bit 6
6	Charge Pump Clock	Output	Clock divided from MCLK (Registers 30 to 31)
7	ADC Data	Input	Enables the calibration function
8	Standard Input	Input	High impedance; level read back through Register 65
9	Input Select	Input	Toggles the input selection (Register 21)
10	Mute All	Input	Logic high forces a mute; low gives normal operation
13	Analog Input Shutdown (default)	Input	Logic high shuts the DAC down
14	Soft Start Complete	Output	High when the DAC output has ramped to ground
15	Output 1	Output	Forced high

Source: [ES9038Q2M Datasheet v1.4 \(ESS Technology\)](#), p. 22 (Register 8). Modes 11 and 12 are reserved.

4. External Components

Ref	Value	Part / Notes	AL-PN
X1	22.5792 MHz	ECS-2520MVLC-225.792-CN-TR, XO, ± 25 ppm, 2.5×2.0 mm, tri-state OE	#1-100
X2	24.576 MHz	Abracon ASTX-H11-24.576MHZ-T, TCXO, ± 2.5 ppm, 3.2×2.5 mm, tri-state OE	#1-101
R_{S1}, R_{S2}	22 Ω , 1%, 0402	YAGEO RC0402FR-0722RL. At each oscillator output; outputs join at X1.	#1-102
R_{EN1}, R_{EN2}	10 k Ω	OE pull-downs. Both oscillators off until firmware enables one.	
R_{RST}	10 k Ω	RESETB pull-down. DAC held in reset until firmware releases it.	
R_{GPIO1}, R_{GPIO2}	10 k Ω	GPIO pull-downs (default mode 13 must stay low)	
$R_{SCL}, R_{SDA}, R_{ADDR}$	10 k Ω	Pull-ups to +3V3_DIG	
C_{DVDD}	4.7 μ F, 6.3 V, X5R, 0402	Murata GRM155R60J475ME47D. Core regulator output (≥ 1 μ F at 1.2 V required).	#1-103
Decoupling	0.1 μ F X7R + 1 μ F X5R	AVCC_L, AVCC_R, both VCCA pins; 0.1 μ F on DVCC and each oscillator VDD	

Oscillator enable: pulling pin 1 low puts the output in high impedance. X1 is disabled at $\leq 0.3 \times VDD$ and enabled at ≥ 0.7

× VDD or open; X2 is disabled below 0.25 × VDD and enabled at ≥ 0.75 × VDD or open. Both parts share one pinout: 1 OE, 2 GND, 3 OUT, 4 VDD.

Source: [ECS-2520MVLC Datasheet \(ECS Inc.\)](#); [ASTX-H11 Datasheet \(Abracon\)](#); [ES9038Q2M Datasheet v1.4 \(ESS Technology\)](#), p. 11 (DVDD capacitor).

5. Equations and Results

Internal MCLK (clock gear divider, Register 0)

$$f_{\text{MCLK}} = \frac{f_{\text{XI}}}{2^{\text{clk_gear}}}, \quad \text{clk_gear} \in \{0, 1, 2, 3\}$$

Source: [ES9038Q2M Datasheet v1.4 \(ESS Technology\)](#), p. 15.

Synchronous mode (DPLL off)

$$f_{\text{MCLK}} = 128 \cdot f_s$$

Source: [ES9038Q2M Datasheet v1.4 \(ESS Technology\)](#), p. 24 (Register 10, 128fs_mode) and p. 52.

Master-mode bit clock (Register 10 master_div default)

$$f_{\text{BCLK}} = \frac{f_{\text{MCLK}}}{2} = 64 \cdot f_s \quad (\text{two 32-bit slots per frame})$$

Source: [ES9038Q2M Datasheet v1.4 \(ESS Technology\)](#), p. 24.

I2C clock limit

$$f_{\text{SCL}} < \frac{f_{\text{MCLK}}}{20} = \frac{5.6448 \text{ MHz}}{20} = 282 \text{ kHz (lowest MCLK)}$$

Source: [ES9038Q2M Datasheet v1.4 \(ESS Technology\)](#), p. 14.

Because clk_gear only divides, each oscillator runs at the highest MCLK its family needs:

Oscillator (XI)	clk_gear	f_s	MCLK	BCLK
22.5792 MHz	÷1 / ÷2 / ÷4	176.4 / 88.2 / 44.1 kHz	22.5792 / 11.2896 / 5.6448 MHz	11.2896 / 5.6448 / 2.8224 MHz
24.576 MHz	÷1 / ÷2 / ÷4	192 / 96 / 48 kHz	24.576 / 12.288 / 6.144 MHz	12.288 / 6.144 / 3.072 MHz

Firmware sequence: hold RESETB low, enable one oscillator, wait for it to start (ECS: ≤ 5 ms), release RESETB, then configure the DAC over I2C. To change family: mute or reset the DAC, swap OE lines, wait, resume.

6. Important Values

Item	Value
X1 phase noise	−159.8 dBc/Hz at 100 kHz offset (typical plot)
X2 phase noise	−158 dBc/Hz at 100 kHz offset
MCLK range	5.6448 to 24.576 MHz

Item	Value
Maximum f_s , synchronous mode	768 kHz (DAC limit)
I2C address	0x92 write (ADDR high)
Power-up state	OE lines, RESETB and GPIO1/2 pulled low: oscillators off, DAC in reset

7. Open Problems

None.

8. Revision Log

Rev	Date	Change
0	2026-09-23	Initial issue: PCM-only interface, pin mapping, clock-source survey, MCLK equations.
1	2026-09-23	Two external oscillators, one per sample-rate family.
2	2026-09-23	XI frequency range recorded.
3	2026-09-23	DAC master, ESP32-S3 slave, DPLL off (synchronous mode).
4	2026-09-23	Oscillator frequencies and clk_gear mapping; I2C speed limit; clock-switching rule.
5	2026-09-23	Oscillators selected: ECS-2520MVLC-225.792-CN-TR, ASTX-H11-24.576MHZ-T.
6	2026-09-23	Pull-downs on both OE lines and RESETB.
7	2026-09-23	$R_{S1}/R_{S2} = 22 \Omega$ at each oscillator output.
8	2026-09-25	Reformatted. Full pin table; GPIO1/GPIO2 functions; $R_{RST} = 10 \text{ k}\Omega$; ESP32-S3 clock sources and jitter ranking removed; equations typeset.
9	2026-09-26	Maximum sample rate closed at 176.4 / 192 kHz (set by oscillator selection).

9. Sources

1. [ES9038Q2M Datasheet v1.4 \(ESS Technology\)](#)
2. [ECS-2520MVLC Datasheet \(ECS Inc.\)](#)
3. [ASTX-H11 Datasheet \(Abracon\)](#)